WORKED SOLUTION

STEP I 1997, Question 2

Pure • Inverse Trigonometric Derivatives

STEP I 1997, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q2

The key idea

In part (i) the derivative of f collapses to zero, so f is constant, but only on each interval where it is defined: the point $x = -1$ splits the real line in two, and the constant is different on each side. In part (ii) it is easier to differentiate x with respect to y and then turn the result upside down; the work is in rewriting $\sin y^2$ and $\cos y^2$ in terms of x and y, which is where the range of y matters.

Solution

Part (i)

The function is defined for every $x \neq -1$. Write $u = \dfrac{1-x}{1+x}$. By the quotient rule,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{-(1+x)-(1-x)}{(1+x)^2} = \frac{-2}{(1+x)^2}, \qquad 1+u^2 = \frac{(1+x)^2+(1-x)^2}{(1+x)^2} = \frac{2(1+x^2)}{(1+x)^2}.$$

Using $\dfrac{\mathrm{d}}{\mathrm{d}x}\tan^{-1} u = \dfrac{1}{1+u^2}\dfrac{\mathrm{d}u}{\mathrm{d}x}$ (chain rule),

$$\begin{aligned} \mathrm{f}'(x) &= \frac{1}{1+x^2} + \frac{(1+x)^2}{2(1+x^2)} \cdot \frac{-2}{(1+x)^2} \\ &= \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0 \qquad (x \neq -1). \end{aligned}$$

Ask yourself Does $\mathrm{f}'(x) = 0$ mean that f is the same constant everywhere?

No. A zero derivative forces a function to be constant only on an *interval* (this comes from the mean value theorem). The domain of f is $(-\infty, -1) \cup (-1, \infty)$, which is two intervals, so f is constant on each piece, and the two constants can differ. We find each one by evaluating at a convenient point, or taking a limit.

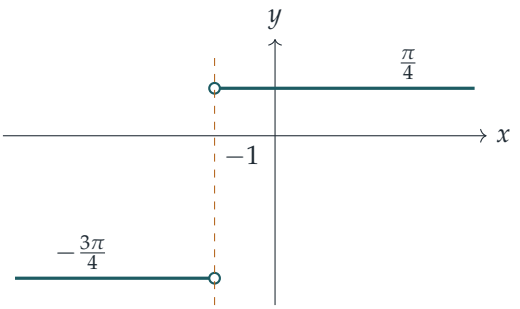
For $x > -1$: put $x = 0$. Then $f(0) = \tan^{-1} 0 + \tan^{-1} 1 = \frac{\pi}{4}$.

For $x < -1$: let $x \to -\infty$. Then $\tan^{-1} x \to -\frac{\pi}{2}$, while $\dfrac{1-x}{1+x} \to -1$, so $\tan^{-1}\left(\dfrac{1-x}{1+x}\right) \to -\frac{\pi}{4}$. Since f is constant on this interval, the constant equals the limit, $-\frac{3\pi}{4}$. (Alternatively, $x = -\sqrt{3}$ gives $\dfrac{1+\sqrt{3}}{1-\sqrt{3}} = -(2+\sqrt{3}) = -\tan\frac{5\pi}{12}$, so $f(-\sqrt{3}) = -\frac{\pi}{3} - \frac{5\pi}{12} = -\frac{3\pi}{4}$.)

Hence

$$f(x) = \begin{cases} \dfrac{\pi}{4}, & x > -1, \\ -\dfrac{3\pi}{4}, & x < -1. \end{cases}$$

The jump of π at $x = -1$ happens because $\dfrac{1-x}{1+x}$ shoots from $-\infty$ to $+\infty$ there, so its inverse tangent leaps from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$.



Part (ii)

Differentiate $x = y \sin y^2$ with respect to y, using the product rule and the chain rule:

$$\frac{dx}{dy} = \sin y^2 + 2y^2 \cos y^2.$$

Now express both trigonometric terms through x and y.

- Since $y > 0$, we may divide by y: $\sin y^2 = \dfrac{x}{y}$.
- Since $0 < y^2 < \frac{\pi}{2}$, we have $\cos y^2 > 0$, so we take the *positive* square root:

$$\cos y^2 = \sqrt{1 - \sin^2 y^2} = \sqrt{1 - \frac{x^2}{y^2}} = \frac{\sqrt{y^2 - x^2}}{y},$$

  where the last step uses $\sqrt{y^2} = y$, true because $y > 0$.

Therefore

$$\frac{dx}{dy} = \frac{x}{y} + 2y\sqrt{y^2 - x^2} = \frac{x + 2y^2\sqrt{y^2 - x^2}}{y}.$$

This is strictly positive, since $x > 0$, $y > 0$ and the square root is non-negative. So we can invert it (the inverse function rule $\frac{dy}{dx} = 1/\frac{dx}{dy}$, valid where $\frac{dx}{dy} \neq 0$):

$$\frac{dy}{dx} = \frac{y}{x + 2y^2\sqrt{y^2 - x^2}},$$

as required.

Ask yourself Why does the question insist on $0 < y < (\pi/2)^{1/2}$?

Without it, $\cos y^2$ could be negative, and the correct formula would have a minus sign in front of the square root. The restriction also makes $y \sin y^2$ strictly increasing on this range (its derivative above is positive), rising from 0 to $(\pi/2)^{1/2}$, so each x in $0 < x < (\pi/2)^{1/2}$ corresponds to exactly one y, and y really is a function of x.

Checks

- Part (i): $f(1) = \frac{\pi}{4} + \tan^{-1} 0 = \frac{\pi}{4}$ and $f(-3) = \tan^{-1}(-3) + \tan^{-1}(-2) \approx -1.2490 - 1.1071 = -2.3562 \approx -\frac{3\pi}{4}$. We checked both constants at 10,000 random points.
- Part (ii): for $x = 0.5$, solving $y \sin y^2 = 0.5$ numerically gives $y \approx 0.81355$, and the formula gives $\frac{dy}{dx} \approx 0.60284$, matching a finite-difference derivative of the numerically inverted function.
- Near $x = 0$, $y \sin y^2 \approx y^3$, so $y \approx x^{1/3}$ with $\frac{dy}{dx} \approx \frac{1}{3y^2}$; the formula gives $y/(y^3 + 2y^3) = \frac{1}{3y^2}$, which agrees.

Common slips

- Concluding $f(x) = \frac{\pi}{4}$ for all x, forgetting that $x = -1$ splits the domain.
- Writing $\cos y^2 = \pm\sqrt{\cdots}$ and not saying why the positive root is the right one.

Beyond STEP

The fact that $f' = 0$ gives a constant only on connected pieces of the domain is a consequence of the mean value theorem, one of the first big results in a university analysis course. Part (ii) is a baby case of the inverse function theorem: a function with non-zero derivative can be inverted locally, and the derivative of the inverse is the reciprocal. Abbott's *Understanding Analysis* treats both carefully.

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